

[3]: *# Create a list with different data types*
`my_list = [10, 3.5, "hello", True]`
Print the list
`print("List:", my_list)`
Print the type of the list
`print("Type of list:", type(my_list))`
Print the type of each element in the list
`for item in my_list:`
 `print(f"Value: {item}, Type: {type(item)}")`



List: [10, 3.5, 'hello', True]
Type of list: <class 'list'>
Value: 10, Type: <class 'int'>
Value: 3.5, Type: <class 'float'>
Value: hello, Type: <class 'str'>
Value: True, Type: <class 'bool'>

```
[4]: # Create two lists
a = [1, 2, 3]
b = [4, 5, 6]
# Concatenation
concat = a + b
print("Concatenation:", concat)
# Repetition (repeat list a 3 times)
repeat = a * 3
print("Repetition:", repeat)
# Membership check (check if 2 is in list a)
print("Is 2 in list a?", 2 in a)
# Length of lists
print("Length of list a:", len(a))
print("Length of list b:", len(b))
```

Concatenation: [1, 2, 3, 4, 5, 6]

Repetition: [1, 2, 3, 1, 2, 3, 1, 2, 3]

Is 2 in list a? True

Length of list a: 3

```
[6]: #Given list
      nums = [10, 5, 20, 8, 15]
      #Find maximum value
      print("Maximum value:",max(nums))
      #Find minimum value
      print("Minimum value:",min(nums))
      #Find sum of elements
      print("Sum of elements:",sum(nums))
      #Sort the list in ascending order
      asc=sorted(nums)
      print("Ascending order:",asc)
      #Sort the list in descending order
      desc = sorted(nums,reverse=True)
      print("Descending order:",desc)
      #Find index of elements 8
      print("Index of 8:", nums.index(8))
```

Maximum value: 20

Minimum value: 5

Sum of elements: 58

Ascending order: [5, 8, 10, 15, 20]

Descending order: [20, 15, 10, 8, 5]

Index of 8: 3

```
[7]: #Create list
lst = [10,20,30]
print("Original list:",lst)
#Add element 40 using append()
lst.append(40)
print("After append:",lst)
#Insert 15 at index 1
lst.insert(1, 15)
print("After insert:",lst)
#Remove element 20
lst.remove(20)
print("After removing 20:",lst)
#Pop last element
lst.pop()
print("After pop:",lst)
#Reverse the list
lst.reverse()
print("After reverse:;",lst)
```

```
Original list: [10, 20, 30]
After append: [10, 20, 30, 40]
After insert: [10, 15, 20, 30, 40]
After removing 20: [10, 15, 30, 40]
After pop: [10, 15, 30]
After reverse:; [30, 15, 10]
```

```
[12]: #Take input from user(space-separated numbers)
      nums = list(map(int, input("Enter numbers separated by space: ").split()))
      #Even numbers
      even = [n for n in nums if n % 2 == 0]
      print("Even numbers:", even)
      #Odd numbers
      odd = [n for n in nums if n % 2 != 0]
      print("Odd numbers:", odd)
      #Largest and smallest number
      print("Largest number:", max(nums))
      print("Smallest number:", min(nums))
      #Bonus: Remove duplicates
      unique_nums = list(set(nums))
      print("List without duplicates:", unique_nums)
```

Enter numbers separated by space: 10 5 20 5 8 10

Even numbers: [10, 20, 8, 10]

Odd numbers: [5, 5]

Largest number: 20

Smallest number: 5

List without duplicates: [8, 10, 20, 5]